

VIRTUAL UNIVERSITY OF PAKISTAN

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
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(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.



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Best of luck!



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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which statement about the General Least Square Method is true?

Answer (Please select your correct option)

- ☐ Solution obtained by this method is always unique.
- ☐ This is a numerical method for the solution of System of Linear Equations.
- ☒ This method find an x that makes Ax as close as possible to the b .
- ☐ This method gives us exact solution of the system.

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

Let $v = (1, -2, 2, 0)$. The unit vector in the same direction as v is

Answer (Please select your correct option)

☐ $\left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}, 0\right)$

☐ $\left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}, 0\right)$

☐ $\left(\frac{-1}{3}, \frac{2}{3}, \frac{-2}{3}, 0\right)$

☐ $\left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right)$

ans

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Let $v = (1, -2, 2, 0)$. The unit vector in the same direction as v has magnitude

Answer (Please select your correct option)

☐ 3

☐ 1

☐ 2

☐ -1

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Let $\mathbb{R}^3$ have the Euclidean inner product. Then $u = (2, 1, 3), v = (1, 7, k)$ are orthogonal for

out of

Answer (Please select your correct option)

☐ $k = 9$

☐ $k = -3$ [check it](#)

☐ $k = -9$

☐ $k = 3$

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

Let $\mathbb{R}^3$ have the Euclidean inner product. Then $u = (k, -3, 1), v = (-3, 5, 6)$ are orthogonal for

out of

Answer (Please select your correct option)

☐ $k = 2$

☐ $k = -2$

☐ $k = 3$

☐ $k = -3$

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

If u and v are orthogonal then

Answer (Please select your correct option)

☐ $\|u - v\|^2 = \|u\| + \|v\|$

☐ $\|u - v\| = \|u\|^2 + \|v\|^2$

☐ $\|u - v\|^2 = \|u\|^2 + \|v\|^2$

☐ $\|u - v\|^{\frac{1}{2}} = \|u\| + \|v\|$

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

Suppose that $A = \begin{bmatrix} 0.5 & 0.6 \\ -0.3 & 1.4 \end{bmatrix}$ has eigenvalues 0.8 and 1.1. Then origin is a

Answer (Please select your correct option)

☐ Saddle point

☐ Repellor

☐ Attractor

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

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Answer (Please select your correct option)

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Question No : 8 of 52

Marks: 1 (Budgeted Time 1 Min)

If A is an $m \times n$ matrix with linearly independent column vectors, then A can be factored as

$$A = QR$$

Where Q is an $m \times n$ matrix with orthonormal column vectors, and R is an $n \times n$

Answer (Please select your correct option)

☐ Upper triangular matrix

ans

not confirm

☐ Invertible matrix

☐ Invertible lower triangular matrix

☐ Invertible upper triangular matrix

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

The QR-Decomposition of a 3×3 matrix A gives

Answer (Please select your correct option)

☐ matrix Q of order 3×1 and R of order 3×1

☐ matrix Q of order 3×3 and R of order 3×1

☐ matrix Q of order 3×3 and R of order 3×3

ans

1 0 0
0 1 0
0 0 1

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

The QR-Decomposition of a 5×2 matrix A gives

Answer (Please select your correct option)

☐ matrix Q of order 5×1 and R of order 5×1

☐ matrix Q of order 5×2 and R of order 2×5

☐ matrix Q of order 5×2 and R of order 2×2

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

If u, v and w are vectors in a real inner product space, and k is any scalar, then which one of the following is not a property for the inner product space

out of

Answer (Please select your correct option)

☐ $\langle 0, v \rangle = \langle v, 0 \rangle$

☐ $\langle u, kv \rangle = k \langle u, v \rangle$

☐ $\langle u, v - w \rangle = \langle u, v \rangle + \langle u, w \rangle$

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

The dominant eigenvalue for the matrix $A = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is

Answer (Please select your correct option)

☐ $\lambda = 1$

☐ $\lambda = -3$

☐ $\lambda = -1$

☐ $\lambda = 0$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ of a matrix is the sum of main diagonal elements of that matrix.

Answer (Please select your correct option)

☐ Trace [check it](#)

☐ Determinent

☐ Eigenvalue

☐ Sum

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

If a square matrix has orthonormal columns, then it also has _____.

Answer (Please select your correct option)

☐ orthonormal rows

☐ orthonormal diagonal

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

If x is orthogonal to both u and v , then x must be _____ to $u - v$.

Answer (Please select your correct option)

☐ orthogonal

☐ orthonormal

☐ perpendicular

☐ parallel

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

If a matrix U has orthonormal columns, then _____ = I

Answer (Please select your correct option)

☐ UU^T

☐ UU

☐ U^{-1}

☐ $U+U^T$

ans

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

If 5 is an eigenvalue of A and x is a corresponding eigenvector, then the eigenvalue of A^2 is

Answer (Please select your correct option)

☐ 5

☐ 25

☐ 10

☐ 15

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$A^2 = 5(2) = 25$

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

The eigenvalues of the matrix $\begin{bmatrix} 3 & -4 & 1 \\ 0 & -5 & 1 \\ 0 & 0 & 4 \end{bmatrix}$ are

we read down to up

Answer (Please select your correct option)

☐ 3,-4,1

☐ 4,-5,3

☐ 3,-5,1

☐ 4,1,-5

ans

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ is diagonalizable, then A is invertible.

Answer (Please select your correct option)

True

ans

False

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

An $n \times n$ matrix A is diagonalizable if and only if A has

Answer (Please select your correct option)

☐ n linearly independent eigenvectors

☐ n^2 linearly independent eigenvectors

☐ $n + 1$ linearly independent eigenvectors

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Let A have eigenvalues 2, 5, 0, -7, and -2. Then the dominant eigenvalue for A is

Answer (Please select your correct option)

☐ $\lambda = 5$

☐ $\lambda = 0$

☐ $\lambda = -7$

☐ $\lambda = 2$

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Which statement is FALSE.

Answer (Please select your correct option)

☐ If $Ax = \lambda x$ for some real number λ then λ is known as eigenvalue of the matrix A.

☐ The eigenvalues of any matrix are on its main diagonal.

☐ In order to find the eigenvalues we solve the equation $|A - \lambda I| = 0$

☐ An eigenspaces of A is the Null space of some matrix.

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

Algebra is a transformation of real life problems into sort of _____.

Answer (Please select your correct option)

☐ logical representation

☐ mathematical representation

☐ physical representation

☐ illogical representation

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

If reduced echelon form of a linear system is $\begin{bmatrix} 1 & 0 & 5 & 5 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ when free variable $x_3 = 0$, then which of the following is true for it?

Answer (Please select your correct option)

☐ The particular solution is (0, 5, 6).

☐ The particular solution is (6, 5, 0).

☐ The particular solution is (5, 6, 0).

☐ The particular solution is (0, 6, 5).

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

If the vector equation $c_1 \vec{v}_1 + c_2 \vec{v}_2 = 0$ with $c_1 = 0 = c_2$ then which of the following is true for $\left\{ \begin{matrix} \vec{r}_1 \\ \vec{v}_1, \vec{v}_2 \end{matrix} \right\}$?

Answer (Please select your correct option)

☐ It is a linearly independent set.

☐ It is a linearly dependent set.

☐ The system of equations is inconsistent.

☐ The system of equations is non- homogeneous.

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is true for the linear operator L defined by $L \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} a_1 \\ -a_2 \end{pmatrix}$?

Answer (Please select your correct option)

☐ It is an enlargement by a negative scale factor.

☐ It is a shear.

☒ It is a reflection about X -axis .

☐ It is a reflection about Y -axis .

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Question No : 27 of 52

Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 4 & -1 \\ 5 & 3 \end{bmatrix}$, then which of the following is the value of $\det(A)$?

Answer (Please select your correct option)

☐ 7

☐ -17

☐ 17

☐ 11

ans

$4(3) - (-1)(5) = 17$

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

If the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$, then which of the following is true about it ?

Answer (Please select your correct option)

☐ Its determinant is 0 .

☐ Its determinant is 1 .

☐ Its determinant is 2 .

☐ Its determinant is 4 .

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

If the matrix $A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 1 & 7 \\ 0 & 0 & 0 \end{bmatrix}$, then which of the following is true about it ?

Answer (Please select your correct option)

☐ Its determinant is 0 .

☐ Its determinant is 2 .

☐ Its determinant is 4 .

☐ Its determinant is 6 .

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If the determinant of the matrix $A = \begin{bmatrix} 4 & 3 & 5 \\ 3 & 1 & 1 \\ 5 & 7 & 7 \end{bmatrix}$ is 32 and the matrix B is obtained by multiplying any row of A with an integer value 4 , then which of the following is true about the matrix B ?

Answer (Please select your correct option)

☐ Its determinant is 18 .

☐ Its determinant is -32 .

☐ Its determinant is 128 .

☐ The information is not sufficient to calculate the determinant .

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

If a set W be a subspace of a vector space V , then which of the following is NOT true for it ?

Answer (Please select your correct option)

- ☐ It must be closed under the scalar multiplication .
- ☐ It may or may not be closed under the operation of addition .
- ☐ It must have an additive inverse of each element .
- ☐ It must be commutative under the operation of addition .

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Let a set S is a basis of a vector space V , then which of the following is NOT true about it ?

Answer (Please select your correct option)

- ☐ It spans V .
- ☐ It is linearly independent.
- ☐ It is linearly dependent.
- ☐ Each element of S belongs to V .

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Question No : 33 of 52

Marks: 1 (Budgeted Time 1 Min)

If a set $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for R^3 and $[\vec{v}]_S = \begin{bmatrix} 7 \\ 5 \\ 3 \end{bmatrix}$, then which of the following is the most appropriate option ?

Answer (Please select your correct option)

☐ $\vec{v} = (1, 0, 0)^t \in R^3$

☐ $\vec{v} = (2, 3, 4)^t \in R^3$

☐ $\vec{v} = (7, 5, 3)^t \in R^3$

☐ $\vec{v} = (2, 7, 4)^t \in R^3$

ans

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

If P is a polynomial of order ' n ', then which of the following is true for the $\dim(P_n)$?

Answer (Please select your correct option)

- ☐ $\dim(P_n) = n - 1$
- ☐ $\dim(P_n) = n$
- ☒ $\dim(P_n) = n + 1$
- ☐ $\dim(P_n) = 1$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Let $A = \begin{bmatrix} 1 & 3 & 2 & -1 & 5 \\ 2 & 2 & 1 & 1 & 4 \\ -1 & 4 & 1 & 2 & -3 \\ 3 & 5 & 3 & -4 & 2 \end{bmatrix}$, the row space of A is the subspace of R^5 . Identify the set by which the row space of A is spanned?

Answer (Please select your correct option)

☐ $\{(1, 2, -1, 3), (2, 4, -2, 6), (2, 1, 1, 3), (-1, 1, 2, -4), (6, 3, 3, 9)\}$

☐ $\{(1, 3, 2, -1, 5), (2, 6, 4, -2, 10), (-1, 4, 1, 2, -3), (-3, 12, 3, 6, -9)\}$

☐ $\{(1, 3, 2, -1, 5), (2, 2, 1, 1, 4), (-1, 4, 1, 2, -3), (3, 5, 3, -4, 2)\}$ [chek it](#)

☐ $\{(1, 2, -1, 3), (3, 2, 4, 5), (2, 1, 1, 3), (-1, 1, 2, -4), (5, 4, -3, 2)\}$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

If $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ and $\vec{x} \in \text{Row}(A)$, then which of the following is the most appropriate option?

Answer (Please select your correct option)

☐ $\vec{x} = c_1(1, 3) + c_2(2, 4)$

☐ $\vec{x} = c_1(1, 2) + c_2(3, 4)$

☐ $\vec{x} = c_1(1, 4) + c_2(3, 2)$

☐ $\vec{x} = c_1(2, 3) + c_2(4, 1)$

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Question No : 37 of 52

Marks: 1 (Budgeted Time 1 Min)

Consider the bases for R^2 given by $B = \{ \vec{b}_1, \vec{b}_2 \}$ and $C = \{ \vec{c}_1, \vec{c}_2 \}$; where $\vec{b}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\vec{c}_1 = \begin{bmatrix} -6 \\ 4 \end{bmatrix}$, $\vec{c}_2 = \begin{bmatrix} -4 \\ 7 \end{bmatrix}$; then which of the following is the change-of-coordinates matrix from C to B ?

Answer (Please select your correct option)

☐ $P_{B \leftarrow C} = \begin{bmatrix} -2 & 3 \\ 4 & 7 \end{bmatrix}$

☐ $P_{B \leftarrow C} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

☐ $P_{B \leftarrow C} = \begin{bmatrix} -6 & -4 \\ 4 & 7 \end{bmatrix}$

☐ $P_{B \leftarrow C} = \begin{bmatrix} -10 & -11 \\ 4 & 7 \end{bmatrix}$

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

If $B = \{\vec{b}_1, \vec{b}_2, \dots, \vec{b}_n\}$ and E is the standard basis $\{e_1, e_2, \dots, e_n\}$ in R^n ; then which of the following is the value of $\begin{bmatrix} 1 \\ \vec{b}_1 \end{bmatrix}_E$?

Answer (Please select your correct option)

☐ E
☐ $\vec{b}_1 + E$
☒ $\vec{b}_1$
☐ $\vec{b}_2$

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\{y_k\} = \{\dots, 1, 0.7, 0, -0.7, -1, -0.7, 0, 0.7, 1, 0.7, 0, \dots\}$ and $0.35y_{k+2} + 0.6y_{k+1} + 0.42y_k = z_k$;
 $\uparrow$
 $k = 0$
 then which of the following is the value of z_0 ?

Answer (Please select your correct option)

☐ 0.840

☐ 0.049

☐ -0.770

☐ -1.139

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